INSTRUCTOR MANUAL

CSE 4571 — Practical Robotics Project with Arduino

Seventh Semester B.Tech (2L – 4P) | Hardware-Integrated Simulation-Based Laboratory

Practical Arduino Robotics — Lukas Kaul

SESSION-1

Ultrasonic Sensor-Based Distance Measurement and Obstacle Detection Using Arduino

Learning Domain Distribution — Cognitive: 45% | Psychomotor: 40% | Affective: 15%

*Note: This is a **Hardware-Integrated Simulation-Based Laboratory** course, where simulation activities are supported by actual hardware components. **Physical components, breadboards, and hardware kits are required** for laboratory activities, along with the prescribed simulation tools.*



SIKSHA 'O' ANUSANDHAN
(Deemed to be) University

1. Session Information

Field	Details
Course Code / Title	CSE 4571 — Practical Robotics Project with Arduino
Semester	Seventh Semester B.Tech (CSE)
Course Format	2L – 4P (Practical / Laboratory, Hardware-Integrated and Simulation-Based)
Session	1
Duration	6 Hours (360 minutes)
Software and Hardware Used	Tinkercad/ Wokwi (Online Circuit Simulation Platform) and physical components
Topic	Ultrasonic Sensor-Based Distance Measurement and Obstacle Detection Using Arduino
Textbook Reference	Practical Arduino Robotics — Lukas Kaul

*Hardware components, apparatus, component kits, and breadboards are integrated where required, while simulation-based activities are performed on-screen using **Tinkercad/Wokwi**.*

2. Learning Objectives

By the end of this session, students will be able to:

Objectives	Statements
O1	Explain the fundamental principle of ultrasonic distance measurement using the HC-SR04 ultrasonic sensor.
O2	Identify the major terminals of the ultrasonic sensor (VCC, Trigger, Echo, and GND) and explain their functions.
O3	Explain how the Arduino UNO generates a trigger pulse, measures the echo time, and converts the measured time into distance using the speed of sound.
O4	Correctly interface the HC-SR04 ultrasonic sensor with the Arduino UNO, ensuring proper signal, power, and common-ground connections.
O5	Implement Arduino-based distance measurement and obstacle detection by defining a threshold distance and generating an appropriate indication when an obstacle is detected.

3. Course Outcomes (CO1–CO6)

CO	Course Outcome Statement	Learning Domain	Taxonomy Level
CO1	Understand the fundamentals of Arduino hardware and software	Cognitive (L) + Psychomotor (P) + Affective (A)	L2 – Understand+ P1: Perception+ A1: Receiving
CO2	Interface various sensors and actuators with Arduino	Cognitive (L) + Psychomotor (P) + Affective (A)	L3 – Apply + P3 – Guided Response+ A2: Responding
CO3	Apply programming logic to control robotic systems.	Cognitive (L) + Psychomotor (P) + Affective (A)	L3 – Apply + P4 – Mechanism + A3: Valuing
CO4	Design and build basic to intermediate level robotics projects.	Cognitive (L) + Psychomotor (P) + Affective (A)	L5 – Evaluate + P5 – Complex Overt Response+ A4: Organization
CO5	Demonstrate problem-solving and debugging skills in robotics.	Cognitive (L) + Psychomotor (P) + Affective (A)	L4 – Analyze + P6 – Adaptation + A4: Organization
CO6	Collaborate in teams to plan, execute, and present robotics projects.	Cognitive (L) + Psychomotor (P) + Affective (A)	L6 – Create + P7 – Origination + A5 – Characterization

L: Cognitive (L1: Remembering; L2: Understanding; L3: Applying; L4: Analyzing; L5: Evaluating; L6: Creating)

P: Psychomotor (P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6: Adaptation; P7: Origination)

A: Affective (A1: Receiving; A2: Responding; A3: Valuing; A4: Organization; A5: Characterization)

The course CSE 4571 - Practical Robotics Project with Arduino primarily reflects the following Sustainable Development Goals (SDGs):

UN Sustainable Development Goals (SDGs)

SDG	Goal	Course Relevance
SDG 4	Quality Education	Develops practical skills in Arduino programming, ultrasonic sensor interfacing, distance measurement, obstacle detection, circuit implementation, simulation, observation, data interpretation, and troubleshooting.
SDG 7	Affordable and Clean Energy	Promotes efficient operation of sensor-based robotic systems by using low-power ultrasonic sensing and Arduino-based control, supporting awareness of energy-efficient embedded and automation systems.
SDG 9	Industry, Innovation and Infrastructure	Promotes innovation in robotic and IoT systems through Arduino-based ultrasonic sensing, real-time distance measurement, obstacle detection, automation, and applications in autonomous robots, smart vehicles, and industrial monitoring systems.
SDG 11	Responsible Consumption and Production	Supports smart and automated systems through obstacle detection and distance sensing, with applications in smart mobility, automated vehicles, intelligent traffic systems, and safer urban environments.

CO – PO Mapping (Focus: CO1-CO6)

CO	PO Mapped	Strength
CO1	PO1, PO2, PO3	High
CO2	PO1, PO2, PO3, PO4, PO5, PO11	High
CO3	PO1, PO2, PO3, PO4, PO5, PO11	High
CO4	PO1, PO2,PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO11	High
CO5	PO1, PO2,PO3, PO4, PO5, PO6, PO7, PO8, PO9,PO10, PO11	High
CO6	PO1, PO2,PO3, PO4, PO5, PO6, PO7, PO8, PO9,PO10, PO11	High

4. Learning Domain Distribution (3H Model)

The 360-minute session strictly maintains the prescribed course-level domain distribution of Cognitive 45%, Psychomotor 40%, and Affective 15%.

Domain	Weightage	Time (approx.)	Focus
Cognitive (Head)	45%	162 minutes	Understanding ultrasonic sensor operation, ultrasonic wave propagation, distance measurement based on echo time, speed of sound, Arduino timing functions, and conversion of echo duration into distance.
Psychomotor (Hand)	40%	144 minutes	Identifying and wiring the Arduino UNO and HC-SR04 ultrasonic sensor; configuring Trigger and Echo pins; running the Arduino program; observing distance readings; implementing threshold-based obstacle detection; simulation/hardware testing and troubleshooting.
Affective (Heart)	15%	54 minutes	Developing safety awareness, careful wiring, systematic observation, responsible handling of electronic components, accurate measurement practices, proper sensor placement, and safe operation of the Arduino-based obstacle detection system.

5. Instructor Preparation

Before opening Tinkercad/Wokwi, the instructor should explain the basic concepts of ultrasonic distance measurement and obstacle detection. Introduce the role of the Arduino UNO as the controller for processing sensor data and making obstacle-detection decisions. Explain the HC-SR04 ultrasonic sensor as the primary input device for measuring the distance of nearby objects.

Briefly explain ultrasonic waves, Trigger and Echo signals, echo time, speed of sound, and time-to-distance conversion in simple terms. Demonstrate the relationship between the transmitted ultrasonic pulse, reflected echo, measured time, and calculated distance. Explain how a predefined distance threshold can be used to identify an obstacle. Physical components may be shown as references, while activities are performed through simulation and integrated hardware, where applicable.

Ultrasonic Sound and Time-of-Flight

Sound above the human hearing limit of about 20 kHz is termed ultrasonic. The HC-SR04 module transmits a short burst of eight cycles at approximately 40 kHz from its transmitting transducer (marked T). The burst travels through air, strikes an obstacle, and a fraction of the energy returns as an echo which is picked up by the receiving transducer (marked R). Because the burst must travel to the target and back, the measured interval represents a round trip; only half of it corresponds to the actual separation. This measurement technique is called time-of-flight (ToF) ranging.

Speed of Sound and the Distance Equation

The speed of sound in dry air at 20 °C is approximately v = 343 m/s = 0.0343 cm/µs. If t is the total round-trip echo time in microseconds, then

$$d = (v \times t) / 2 = (0.0343 \times t) / 2 = t / 58.3 \approx t / 58 \quad \text{(cm)}$$

which is exactly the divisor used in the program. The speed of sound rises by roughly 0.6 m/s per °C, so measurements taken in a warm laboratory read slightly short unless the constant is temperature-compensated as v = 331.4 + 0.6T m/s.

Worked example: An echo pulse of 1160 µs gives d = 1160/58 ≈ 20.0 cm, which falls inside the WARNING zone of this experiment.

HC-SR04 Trigger and Echo Timing

The module is controlled by a strict handshake. The Arduino holds TRIG LOW to settle the line, raises it HIGH for at least 10 µs, and lowers it again. The module then emits the burst automatically and drives ECHO HIGH for a period equal to the round-trip time. The relationship is shown in Figure 1.

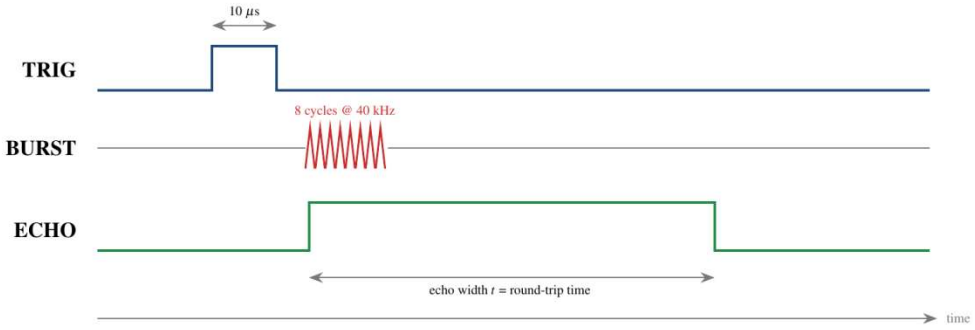


Figure 1. HC-SR04 handshake: a 10 µs trigger pulse initiates the 40 kHz burst; the ECHO line then stays HIGH for the round-trip travel time.

Measuring the Echo Width with pulseIn()

The Arduino function pulseIn(pin, HIGH, timeout) blocks until the named pin goes HIGH, times the HIGH interval, and returns its width in microseconds. It is used here because the echo width is far too short — tens to thousands of microseconds — to be resolved by repeated digitalRead() polling inside a loop. The third argument sets a timeout of 30 000 µs; if no echo returns within that window the function returns 0, and the program reports the reading as out of range instead of freezing. The useful range of the HC-SR04 is roughly 2 cm to 400 cm, with a resolution of about 3 mm and a beam angle near 15°.

Threshold-Based Decision Logic

Rather than reacting to a raw number, the controller classifies the measurement into three decision zones. This is the simplest form of the reactive behaviour used by obstacle-avoiding robots.



Figure 2. Threshold zone

LED Indicators and the Active Buzzer

An LED conducts only in forward bias and drops about 2 V. Connected directly to a 5 V pin it would draw a destructive current, so a series resistor is mandatory:

$$R = (V_{pin} - V_{LED}) / I_{LED} = (5 - 2) / 0.013 \approx 230\ \Omega \Rightarrow \text{use } 220\ \Omega \text{ (standard value)}$$

giving roughly 13–14 mA, comfortably within the 20 mA per-pin limit of the ATmega328P. An active buzzer contains its own oscillator and sounds whenever it is energised; a passive buzzer requires a driving frequency, which is what tone(pin, 1000) supplies. tone() with a third argument produces a beep of fixed duration, while noTone() silences the pin.

System Block Diagram

The experiment realises the canonical embedded chain that every robot repeats: sense, decide, act.

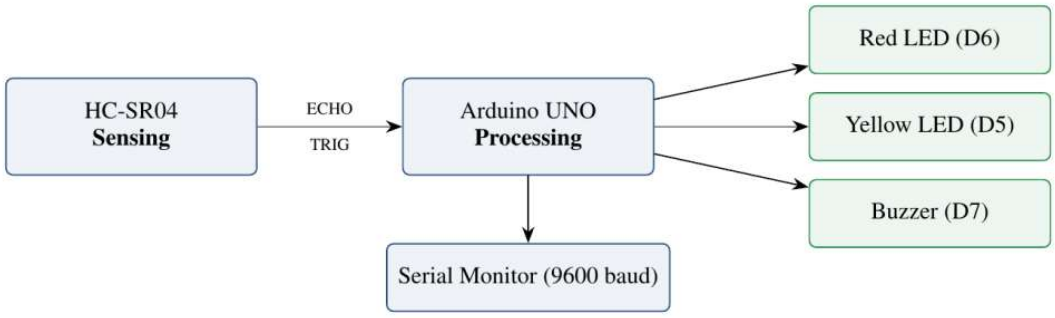


Figure 3. Sensor–controller–actuator signal flow

Objectives

- Understand ultrasonic time-of-flight distance measurement.
- Interface a digital distance sensor with the Arduino UNO.
- Acquire the echo duration and convert it into distance.
- Implement threshold-based decision making.
- Use LED and buzzer outputs as visual and audible alerts.
- Develop confidence in basic sensor–controller–actuator integration.

Components Required (Tinkercad/Wokwi and Hardware)

No.	Component	Quantity	Purpose
1	Arduino UNO	1	Microcontroller board; acquires the echo and drives all outputs
2	HC-SR04 Ultrasonic Sensor	1	Time-of-flight distance measurement
3	Red LED	1	DANGER zone indication
4	Yellow LED	1	WARNING zone indication
5	Active Buzzer	1	Audible alert
6	220 Ω Resistor	2	LED current limiting
7	Breadboard	1	Circuit assembly
8	Jumper Wires	As required	Interconnections
9	USB Cable / Wokwi account	1	Programming, power and Serial Monitor access

Arduino Program

```
const int TRIG = 9;
const int ECHO = 10;
const int RED_LED = 6;
const int YELLOW_LED = 5;
const int BUZZER = 7;

float getDistance() {
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
```

```
  long duration = pulseIn(ECHO, HIGH, 30000);

  if (duration == 0)
    return -1;  // Timeout

  return duration / 58.0;  // Distance in cm
}

void setup() {
  Serial.begin(9600);

  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(YELLOW_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);
}

void loop() {
  float d = getDistance();

  if (d < 0) {
    Serial.println("No echo / out of range");
    digitalWrite(RED_LED, LOW);
    digitalWrite(YELLOW_LED, LOW);
    noTone(BUZZER);
  }

  else if (d <= 10) {
    Serial.print(d);
    Serial.println(" cm -> DANGER");

    digitalWrite(RED_LED, HIGH);
    digitalWrite(YELLOW_LED, LOW);
    tone(BUZZER, 1000);
  }

  else if (d <= 25) {
    Serial.print(d);
    Serial.println(" cm -> WARNING");

    digitalWrite(RED_LED, LOW);
    digitalWrite(YELLOW_LED, HIGH);
    tone(BUZZER, 1000, 150);
  }

  else {
    Serial.print(d);
    Serial.println(" cm -> SAFE");

    digitalWrite(RED_LED, LOW);
    digitalWrite(YELLOW_LED, LOW);
    noTone(BUZZER);
  }

  delay(250);
}
```

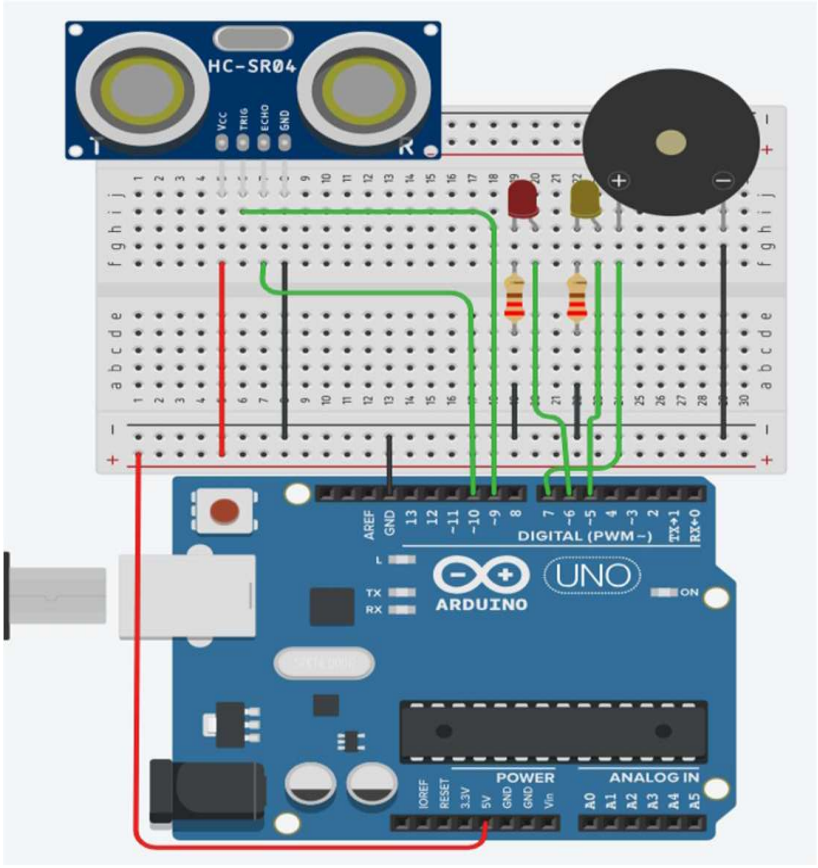


Figure 4. Arduino UNO–HC-SR04 breadboard wiring with LED and buzzer indicators.

Figure 5. Hardware setup for Arduino UNO–HC-SR04 breadboard wiring with LED and buzzer indicators.

6. Software & Resources Required

Resource	Purpose
Tinkercad (www.tinkercad.com)	Primary simulation platform for circuit design, Arduino programming, and ultrasonic sensor-based distance measurement and obstacle detection activities.
Wokwi (wokwi.com)	Alternative simulation platform for Arduino-based circuit design, programming, and testing of HC-SR04 ultrasonic sensor distance measurement and obstacle detection.
Web Browser (Chrome/Edge, updated)	Tinkercad and Wokwi run directly in the browser; no software installation is required.
Projector / Screen Share	Enables live instructor demonstrations of circuit wiring, programming, sensor operation, distance measurement, and obstacle detection.
Observation Sheet	Used by students to record Trigger and Echo signals, measured distances, obstacle detection status, observations, and experimental outcomes.

*This experiment follows a **Hardware-Integrated and Simulation-Based** approach. Physical components, kits, and Arduino-based hardware are used for hands-on implementation and verification, while **Tinkercad /Wokwi** are used for circuit design, programming, simulation, and Ultrasonic Sensor-Based Distance Measurement and Obstacle Detection testing.*

7. Student Preparation (Pre-Lab)

- Create a free **Tinkercad/Wokwi** account prior to the session or join the instructor's Tinkercad/Wokwi Classroom using the provided class code.
- Review the pre-lab material on **ultrasonic sensors, Arduino UNO, distance measurement, obstacle detection, LEDs, and buzzer-based alerts** before the session.
- Bring a notebook to record circuit connections, measured distances, obstacle detection results, LED/buzzer status, observations, and experimental results.

8. Minute-wise Instructor & Student Activity Plan

Time	Instructor Activity	Student Activity	Domain
0–20 min	Introduce the experiment, aim, objectives, laboratory/simulation practices, safety requirements, and applications of ultrasonic distance measurement and obstacle detection in robotics.	Listen to the introduction, understand the importance of safe laboratory/simulation practices, and show readiness to participate in the activity.	**Affective (Heart)**
20–60 min	Explain the working principle of the ultrasonic sensor, including trigger signal, echo signal, time-of-flight measurement, and distance calculation. Discuss practical applications of obstacle detection.	Understand the operating principle of the ultrasonic sensor and explain its role in obstacle detection, autonomous robots, smart vehicles, and automation systems.	**Cognitive (Head)**
60–95 min	Explain the Arduino UNO, ultrasonic sensor, LEDs, buzzer, digital input/output pins, and the relationship between echo time and measured distance.	Identify the components and explain the function of each component in the obstacle detection system.	**Cognitive (Head)**
95–135 min	Demonstrate Arduino UNO, HC-SR04 ultrasonic sensor, LEDs, buzzer, and breadboard identification. Show VCC, GND, TRIG, ECHO, LED, and buzzer connections in Tinkercad/Wokwi or hardware.	Identify the components and terminals, arrange the components, build the circuit, and verify the required power, signal, and common-ground connections.	**Psychomotor (Hand)**
135–180 min	Explain the Arduino program using digitalWrite(), pulseIn(), distance calculation, Serial Monitor, and conditional statements for SAFE, WARNING, and DANGER zones.	Interpret the program statements and explain the complete sequence: ultrasonic pulse → echo detection → time measurement → distance calculation → obstacle classification.	**Cognitive (Head)**

180–220 min	Demonstrate program entry/upload and guide students in running the ultrasonic distance measurement and obstacle detection program in simulation or hardware.	Enter/upload the Arduino program, start the simulation or hardware test, place objects at different distances, and verify the sensor response.	**Psychomotor (Hand)**
220–260 min	Guide systematic experimentation using different obstacle distances and demonstrate the use of the Serial Monitor for recording measured distance and detection status.	Place obstacles at different distances and record at least five distance readings along with the corresponding **SAFE, WARNING, or DANGER** status and LED/buzzer response.	**Psychomotor (Hand)**
260–302 min	Explain the distance thresholds (≤10 cm for DANGER, 10–25 cm for WARNING, and >25 cm for SAFE), timeout conditions, sensor limitations, measurement deviations, and troubleshooting logic.	Analyze the effect of obstacle distance on system response, compare expected and measured values, and identify possible causes of errors involving wiring, sensor alignment, code, power, or environmental conditions.	**Cognitive (Head)**
302–326 min	Guide students in modifying the distance thresholds and testing different obstacle-detection conditions in simulation or hardware.	Modify the program, test different threshold values, verify LED and buzzer responses, and correct simple wiring or programming errors.	**Psychomotor (Hand)**
326–360 min	Discuss safe operation, responsible handling of electronic components, teamwork, and practical applications of ultrasonic sensing. Conduct review/Q&A and provide post-lab instructions.	Participate in discussion and Q&A, demonstrate safety awareness and teamwork, summarize the experiment, and note the post-lab work.	**Affective (Heart)**

9. Instructor Demonstration Steps (Simulation and Hardware Implementation)

Step	Procedure
Step 1	Sign in to **Tinkercad** and create a new Arduino circuit/project.
Step 2	Add an **Arduino UNO, HC-SR04 ultrasonic sensor, red LED, yellow LED, buzzer, breadboard, and jumper wires** to the workspace.
Step 3	Connect the **HC-SR04** to **5V, GND, TRIG (D9), and ECHO (D10)**. Connect the **red LED to D6**, **yellow LED to D5**, and **buzzer to D7**, with appropriate LED resistors and a common ground.
Step 4	Enter the Arduino program to generate the ultrasonic trigger pulse, read the echo duration using `pulseIn()`, and calculate the distance in centimeters.
Step 5	Start the simulation and place/move an obstacle at different distances. Observe how the **measured distance and obstacle status** change.
Step 6	Open the **Serial Monitor** and observe the displayed **distance value and SAFE, WARNING, or DANGER status**.
Step 7	Test the programmed thresholds: **≤10 cm – DANGER**, **10–25 cm – WARNING**, and **>25 cm – SAFE**. Observe the corresponding **red LED, yellow LED, and buzzer** responses.
Step 8	Stop the simulation, verify the circuit and program, save the project, and rename it as **Ultrasonic_Obstacle_Detection_[InstructorName]**.
Step 9	**Repeat Steps 2–8 using hardware components** for the physical implementation of the experiment.

10. Guided / Probing Questions

- Modify the program so that the DANGER threshold is changed from 10 cm to 15 cm. What changes in the system response?
- Why is the TRIG pin of the ultrasonic sensor connected to an Arduino digital output pin?
- Why is the ECHO pin of the ultrasonic sensor connected to an Arduino digital input pin?
- Write the Arduino code required to display both the measured distance and obstacle status on the Serial Monitor.
- What does the pulseIn() function do in the ultrasonic distance-measurement program?

- Modify the program so that the system operates with the following conditions: ≤15 cm – DANGER, 15–30 cm – WARNING, and >30 cm – SAFE.
- What is the purpose of the duration / 58.0 calculation in the program?
- The ultrasonic sensor is connected to TRIG = D9 and ECHO = D10, but the program uses different pin numbers. What program modification is required, and what happens if it is not corrected?
- The distance value is not displayed correctly in the Serial Monitor. Identify possible hardware and software connection faults.
- The Serial Monitor displays "No echo / out of range." What could be the possible reasons for this condition?
- An obstacle is placed at approximately 10 cm from the sensor. What obstacle status should be displayed, and which LED and buzzer response should occur?
- How would you modify the program to change the WARNING range from 10–25 cm to 15–40 cm?
- Why is distance-threshold classification important in an obstacle-detection system?
- Explain why the Arduino, ultrasonic sensor, LEDs, and buzzer must share a common ground.
- The measured distance fluctuates significantly even when the obstacle is stationary. How would you troubleshoot sensor alignment, wiring, power supply, and environmental factors?
- How would you modify the hardware circuit to use two ultrasonic sensors simultaneously with one Arduino for detecting obstacles in different directions?
- What problems can occur if the ultrasonic sensor is incorrectly powered or wired, and how can these problems be avoided?
- Suggest one practical robotic application in which an ultrasonic sensor can be used for distance measurement or obstacle detection.

11. Hands-on Activity (Student Task)

Task: Build and simulate an Ultrasonic Sensor-Based Distance Measurement and Obstacle Detection System using Tinkercad/Wokwi and Hardware.

- Log in and create a project named Ultrasonic_Obstacle_Detection_[YourName]_[RollNo].
- Place an Arduino UNO, HC-SR04 ultrasonic sensor, red LED, yellow LED, buzzer, breadboard, resistors, and jumper wires, and complete the required circuit connections.
- Run the Arduino program to generate the ultrasonic trigger pulse, measure the echo duration, calculate the distance in centimeters, and classify the obstacle as SAFE, WARNING, or DANGER.
- Implement the hardware circuit with proper precautions while handling the Arduino, ultrasonic sensor, LEDs, buzzer, and electrical connections.
- Record at least five observations of obstacle position, measured distance, obstacle status, LED indication, and buzzer response. Verify the programmed thresholds of ≤10 cm (DANGER), 10–25 cm (WARNING), and >25 cm (SAFE).
- Save the project and keep the Tinkercad/Wokwi project link ready for instructor review.
- Take a snapshot/video of the working hardware demonstrating distance measurement and obstacle detection for instructor review.

12. Discussion Points

- **Ultrasonic Sensor Applications in Robotics** – Discuss the use of ultrasonic sensors for distance measurement and obstacle detection in autonomous robots, robotic vehicles, smart parking systems, and automated navigation.
- **Distance-Based Obstacle Detection Concept** – Discuss how the ultrasonic sensor measures the time taken by an ultrasonic pulse to return as an echo, which the Arduino converts into distance and classifies into **SAFE, WARNING, or DANGER** conditions.

- Simulation, Hardware Integration, and Safe Operation – Discuss the use of Tinkercad/Wokwi for simulation and testing, along with proper sensor wiring, LED and buzzer connections, systematic observations, precautions while handling hardware circuits, and organized project management.

13. Reflection Questions (End of Session)

- What is one new concept you learned about ultrasonic distance measurement and obstacle detection today?
- Which part of the Tinkercad/Wokwi simulation did you find most intuitive, and which part was confusing?
- How confident do you feel about explaining the relationship between ultrasonic echo time, measured distance, and SAFE/WARNING/DANGER status (rate 1–5), and why?

14. Home Assignment

- Modify the **Tinkercad/Wokwi ultrasonic sensor circuit** and record five sets of **obstacle distance, measured distance, and corresponding detection status (SAFE/WARNING/DANGER)**.
- Modify the Arduino program to change the **distance detection thresholds** and verify the obstacle-detection operation through simulation.
- Write a short note explaining one practical **robotic application of ultrasonic-based obstacle detection**, such as an autonomous robot, robotic vehicle, smart parking system, or collision-avoidance system.

15. Assessment Rubrics

Rating Scale: 4 = Excellent, 3 = Good, 2 = Satisfactory, 1 = Needs Improvement (aligned to course domain weightage: Cognitive 45% / Psychomotor 40% / Affective 15%)

Criterion	Domain	4 – Excellent	3 – Good	2 – Satisfactory	1 – Needs Improvement
Component Identification & Circuit Construction	Cognitive / Psychomotor	Correctly identifies all components and independently builds the ultrasonic sensor, LED, and buzzer circuit with accurate connections.	Identifies most components and builds the circuit with minor connection errors.	Identifies some components and requires assistance with circuit construction.	Unable to identify components or construct the circuit correctly.
Distance Measurement & Obstacle Detection	Cognitive / Psychomotor	Accurately measures distances, correctly identifies SAFE/WARNING/DANGER conditions, records observations, and clearly explains the sensor operation.	Performs distance measurement and obstacle detection correctly with minor errors and records most observations accurately.	Performs the experiment with assistance and records limited or partially correct observations.	Unable to perform distance measurement or obstacle detection correctly.
Programming & Simulation	Cognitive / Psychomotor	Correctly implements the Arduino program, distance calculation, threshold-based detection, and simulation; independently troubleshoots errors.	Runs the program and simulation correctly with minor errors or assistance.	Requires considerable assistance with programming, simulation, or threshold modification.	Unable to run, modify, or troubleshoot the program or simulation.

Laboratory Engagement & Teamwork	Affective	Participates actively, follows laboratory practices, maintains an organized workspace, handles hardware safely, and supports peers.	Participates regularly, follows safety practices, and cooperates well with peers.	Participates when prompted and demonstrates limited teamwork or safety awareness.	Remains passive, disengaged, or does not follow laboratory practices or cooperate with peers.

16. Student Observation Sheet (Template)

Sl. No.	Actual Distance (cm)	Measured Distance (cm)	Error (cm)	Zone	Buzzer State
1					
2					
3					
4					
5					

Expected Output / Decision Logic

Distance	Indicator	Expected Behaviour
0–10 cm	DANGER	Red LED ON + continuous buzzer
10–25 cm	WARNING	Yellow LED ON + short buzzer tone
>25 cm	SAFE	Both LEDs OFF + buzzer OFF

17. Expected Learning Outcome

At the end of the session, a student should be able to independently open Tinkercad/Wokwi, construct and simulate an ultrasonic sensor-based distance measurement and obstacle detection circuit using Arduino UNO, correctly program the ultrasonic sensor for distance calculation, detect obstacles at different distances, classify them into SAFE, WARNING, and DANGER zones, record the corresponding distance measurements and system responses, and explain in their own words how ultrasonic-based obstacle detection is used in robotic and automation applications.

18. Instructor Tips

- Keep the initial explanation concise and focused on ultrasonic sensor operation, distance measurement, Arduino, echo-time calculation, and obstacle-detection logic.
- Closely monitor students during the hands-on activity, particularly while making Arduino–ultrasonic sensor–LED–buzzer connections and troubleshooting wiring, sensor, or programming errors.
- Encourage peer-assisted learning and emphasize the relationship between obstacle position → ultrasonic echo time → measured distance → detection zone → LED/buzzer response.
- Maintain the Hardware-Integrated and Simulation-Based approach, using physical hardware for hands-on verification and Tinkercad/Wokwi for circuit design, programming, simulation, and testing.

CSE 4571 — Practical Robotics Project with Arduino

Basic Questions

No.	Question	Learning Domain	Taxonomy Level
1	What is an ultrasonic sensor?	Cognitive	Bloom's Taxonomy – Remember
2	Name any two applications of ultrasonic sensors in robotics.	Cognitive	Bloom's Taxonomy – Remember
3	What is meant by obstacle detection?	Cognitive	Bloom's Taxonomy – Understand
4	What is the role of an Arduino UNO in ultrasonic distance measurement?	Cognitive	Bloom's Taxonomy – Understand
5	What is the working principle of the HC-SR04 ultrasonic sensor?	Cognitive	Bloom's Taxonomy – Understand
6	What are the functions of the TRIG and ECHO pins of an ultrasonic sensor?	Cognitive	Bloom's Taxonomy – Remember
7	Which Arduino pins are used for TRIG and ECHO in this experiment?	Cognitive	Bloom's Taxonomy – Remember
8	What is the purpose of the pulseIn() function in the Arduino program?	Cognitive	Bloom's Taxonomy – Understand
9	How is the distance calculated from the measured echo duration?	Cognitive	Bloom's Taxonomy – Understand
10	What is the purpose of duration / 58.0 in the program?	Cognitive	Bloom's Taxonomy – Understand
11	What happens when an obstacle is placed very close to the ultrasonic sensor?	Cognitive	Bloom's Taxonomy – Analyze
12	Why is the ultrasonic sensor connected to digital pins rather than an analog input in this experiment?	Cognitive	Bloom's Taxonomy – Understand
13	What is the purpose of the if–else conditions used in the obstacle-detection program?	Cognitive	Bloom's Taxonomy – Understand
14	What detection zones are defined in the experiment?	Cognitive	Bloom's Taxonomy – Remember
15	Calculate the approximate distance when the measured echo duration is 580 µs.	Cognitive	Bloom's Taxonomy – Apply
16	An ultrasonic sensor measures an echo duration of 1160 µs. Calculate the approximate distance in centimeters.	Cognitive	Bloom's Taxonomy – Apply
17	If an obstacle is measured at 8 cm, which zone is detected and what LED/buzzer response is expected?	Cognitive	Bloom's Taxonomy – Apply
18	If the measured distance is 20 cm, determine the detection zone and expected output.	Cognitive	Bloom's Taxonomy – Apply
19	If the measured distance is 35 cm, determine the detection zone and expected LED and buzzer states.	Cognitive	Bloom's Taxonomy – Apply
20	The actual distance is 20 cm and the measured distance is 22 cm. Calculate the measurement error.	Cognitive	Bloom's Taxonomy – Apply
21	If the actual distance is 30 cm and the measured distance is 28 cm, calculate the absolute error.	Cognitive	Bloom's Taxonomy – Apply

22	Write the Arduino statement required to generate a 10-microsecond trigger pulse for the ultrasonic sensor.	Cognitive	Bloom's Taxonomy – Apply
23	Write the Arduino statement required to read the echo duration using pulseIn().	Cognitive	Bloom's Taxonomy – Apply
24	Modify the program so that the DANGER zone is changed from ≤10 cm to ≤15 cm.	Cognitive	Bloom's Taxonomy – Apply
25	Modify the program so that the WARNING range becomes 15–30 cm and SAFE is defined as >30 cm.	Cognitive	Bloom's Taxonomy – Apply
26	What happens if the TRIG and ECHO pin numbers in the program do not match the actual circuit connections?	Cognitive	Bloom's Taxonomy – Analyze
27	The Serial Monitor displays "No echo / out of range." Identify two possible causes and suggest solutions.	Cognitive	Bloom's Taxonomy – Analyze
28	The measured distance fluctuates continuously even though the obstacle is stationary. What possible hardware or environmental factors could cause this problem?	Cognitive	Bloom's Taxonomy – Analyze
29	Why must the Arduino, ultrasonic sensor, LEDs, and buzzer share a common ground?	Cognitive	Bloom's Taxonomy – Understand
30	Why is simulation useful before implementing the ultrasonic obstacle-detection circuit on physical hardware?	Cognitive	Bloom's Taxonomy – Understand
31	Can you locate the Circuits option in Tinkercad or create a new Arduino project in Wokwi?	Psychomotor	Simpson's Taxonomy – Perception
32	Can you identify the Arduino UNO, HC-SR04 sensor, LEDs, buzzer, breadboard, workspace, and simulation controls?	Psychomotor	Simpson's Taxonomy – Perception
33	Can you search for and place the Arduino UNO, ultrasonic sensor, LEDs, buzzer, and breadboard in the workspace?	Psychomotor	Simpson's Taxonomy – Set
34	Can you arrange the Arduino, ultrasonic sensor, LEDs, and buzzer correctly in the workspace?	Psychomotor	Simpson's Taxonomy – Guided Response
35	Can you identify the VCC, GND, TRIG, ECHO, LED, and buzzer connections required for the experiment?	Psychomotor	Simpson's Taxonomy – Perception
36	Can you correctly connect the TRIG and ECHO pins of the ultrasonic sensor to the Arduino?	Psychomotor	Simpson's Taxonomy – Guided Response
37	Can you enter, compile, and upload the Arduino program in Tinkercad/Wokwi without assistance?	Psychomotor	Simpson's Taxonomy – Mechanism
38	Can you place an obstacle at different distances and verify the corresponding SAFE, WARNING, and DANGER responses?	Psychomotor	Simpson's Taxonomy – Complex Overt Response
39	Can you modify the Arduino program to change the distance thresholds and verify the resulting system behaviour?	Psychomotor	Simpson's Taxonomy – Adaptation
40	Can you independently troubleshoot incorrect sensor readings, LED indications, or buzzer responses and make the required corrections?	Psychomotor	Simpson's Taxonomy – Origination